

Magnetometer-free Realtime Inertial Motion Tracking by Exploitation of Kinematic Constraints in 2-DoF Joints

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Abstract—Inertial Measurement Units (IMUs) are used to track the motion of kinematic chains in a wide variety of robotic and biomedical applications. However, inertial motion tracking is severely limited by the fact that magnetic fields are inhomogeneous in indoor environments and near electronic devices. Methods that use only accelerations and angular rates for orientation estimation yield no absolute heading information and suffer from heading drift. To overcome this limitation, we propose a novel method that exploits an orientation-based kinematic constraint in joints with two degrees of freedom (DoF), such as cardan joints, saddle joints, the human wrists, elbow or ankles. The method determines the relative heading of the joint segments in real time by minimization of a nonlinear cost function. A filter for singularity treatment ensures accurate tracking during motion phases for which the cost function minimum is ambiguous. We experimentally validate the method in metacarpophalangeal (MCP) joints between the palm and the fingers. Accurate relative orientation tracking is achieved continuously despite several singular motion phases and even though the heading components of the 6D orientations drift by more than 360 degrees within ten minutes. The proposed method overcomes a major limitation of inertial motion tracking and thereby facilitates the use of this technology in robotic and biomechanical applications.

I. INTRODUCTION

In various robotic and biomechanical applications, such as rehabilitation robotics, feedback-controlled neuroprosthetics or human motion assessment, motion tracking plays a central role and is nowadays increasingly performed using inertial sensors [1], [2], [3], [4]. In contrast to marker-based optical motion tracking systems, inertial sensors are completely wearable and neither require an elaborate camera setup nor a clear line of sight.

Inertial and magnetic sensors, also known as 9D inertial measurement units (IMUs), comprise 3D accelerometers, 3D gyroscopes and 3D magnetometers. Sensor fusion of these measurements yields the complete orientation of the sensor with respect to a fixed inertial frame, which is crucial for determining the velocity and position of the sensor as well as the joint angles between neighboring segments in robotic or biomechanical limbs. However, 9D sensor fusion only yields accurate orientation estimates if the magnetic field is homogeneous. Magnetometer readings are known to be highly unreliable in indoor environments and near ferromagnetic material and electronic devices. Abundant research

shows that inside buildings the local magnetic field vector may easily vary by more than a factor two in less than 20 centimeters [5], [6], [7], [8]. Therefore, 9D sensor fusion fails in numerous environments that are relevant for most robotic and biomechanical applications [9].

In 6D inertial sensor fusion, the magnetometer readings are omitted, and the orientations of the sensors are determined from the measured accelerations and angular rates. The heading of such an orientation exhibits an unknown and arbitrary offset, i.e. there is no absolute heading information.

This can be overcome by a calibration pose ensuring precise initial alignment of the sensors at the beginning of a measurement. As shown in [9], this approach leads to improved results compared to conventional 9D sensor fusion, especially in realistic indoor environments. However, the heading still slowly drifts due to integration of gyroscope bias around the vertical axis. Even with very precise bias calibration, this drift easily reaches 90 degrees and more within a few minutes. This renders it impossible to determine long-time stable joint angles from 6D sensor orientations.

In recent literature it has been shown that the missing relative heading information can be obtained by exploiting kinematic models and constraints of different joint types.

(1) For joints with one major degree of freedom (DoF) such as the knee joint, the joint angle can be calculated without the use of magnetometers [10], [11]. Moreover, the relative heading information can be determined in real time by using an orientation-based constraint [12], which facilitates 3D visualization of the body segments.

(2) For rotational joints with two degrees of freedom such as the elbow or wrist, an angular-rates-based constraint

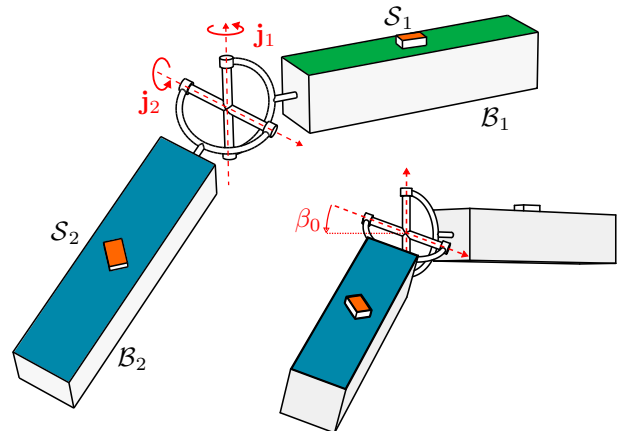


Fig. 1. Kinematic model of a 2-DoF joint with segments B_1 , B_2 , joint axes j_1 , j_2 and inertial sensors S_1 , S_2 . *Right:* View in the j_1 - j_2 -plane. The angle β_0 describes the deviation of the angle between j_1 and j_2 from 90° .

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was used to perform sensor-to-segment calibration and simultaneously identify the relative heading of both segments [13]. The limited degrees of freedom in the relative segment orientation of the elbow and the hand have been exploited in a magnetometer-free method that requires initial sensor-to-segment calibration protocols [14], [15].

(3) In joints with up to three rotational degrees of freedom, position-based constraints can be formulated in models that use the orientation, velocity and position of each segment as states and assume the sensor-to-joint positions to be known. This approach was realized in an optimization-based method for offline motion analysis [16], which also allows for additional hinge joint constraints. A similar but online-capable method, which uses a sliding-window approach, is described in [17]. However, both [16] and [17] still require the use of magnetometers for the initial orientation. Another approach based on an extended Kalman filter (EKF) is proposed in [18], but it has not yet been validated in biological joints.

Experimental validation in previous studies has almost exclusively been carried out under idealistic assumptions of either (1) rigid mechanical setups with perfect joint kinematics, (2) short time durations without sufficient focus on long-time stability, or (3) motions that persistently excite the degrees of freedom of the joints and thus yield sufficient relative-heading information at all times. Motions during which the joint segments remain close to singularities of the constraints (i.e. poses with unobservable relative heading) for more than a few seconds have barely been considered or discussed in previous studies.

In the present paper, we propose a novel magnetometer-free method for relative realtime motion tracking in 2-DoF joints. A new, orientation-based constraint is proposed and exploited to determine the relative heading by window-based online optimization. In contrast to previous approaches, the proposed method requires neither an initial-pose calibration nor sensor-to-joint position parameters, its long-time stability is validated experimentally in the metacarpophalangeal joints (MCP) of the hand, and its advantage over a conventional magnetometer-free method is demonstrated.

II. METHOD

Two segments $\mathcal{B}_1$ and $\mathcal{B}_2$ are connected by a joint with two degrees of freedom, which can be modeled as a kinematic chain of two hinge joints as shown in Fig. 1. Denote the joint axes as $\mathbf{j}_1, \mathbf{j}_2 \in \mathbb{R}^3$, with Euclidean norm $\|\mathbf{j}_1\| = \|\mathbf{j}_2\| = 1$. The coordinates of $\mathbf{j}_1$ are fixed in the frame $\mathcal{B}_1$ and the coordinates of $\mathbf{j}_2$ are fixed in the frame $\mathcal{B}_2$. Without loss of generality, define the coordinate systems $\mathcal{B}_1$ and $\mathcal{B}_2$ such that $\mathbf{j}_1 = [0 \ 0 \ 1]^T$ and $\mathbf{j}_2 = [0 \ 1 \ 0]^T$.

The angle between $\mathbf{j}_1$ and $\mathbf{j}_2$ is fixed, but not necessarily 90° . As shown in Fig. 1, we use β_0 to denote the deviation of this angle from 90° .¹ This kinematic model is general enough to describe many different joints such as saddle joints, Cardan joints, the human wrist, elbow, or the ankle.

¹For the elbow joint, this angle β_0 is commonly called carrying angle and approximately $5\text{--}15^\circ$, see for example [19].

Two inertial sensors $\mathcal{S}_1$ and $\mathcal{S}_2$ are placed on the segments in known orientation, i.e. the orientations of the sensors w.r.t. the segments $\mathcal{S}_1^1 \mathbf{q}$ and $\mathcal{S}_2^2 \mathbf{q}$ are known. This can be achieved by precise attachment or by using automatic sensor-to-segment calibration methods [13], [20]. The sensors measure the angular rates $\boldsymbol{\omega}_1(t)$ and $\boldsymbol{\omega}_2(t)$ as well as the accelerations $\mathbf{a}_1(t)$ and $\mathbf{a}_2(t)$ in local coordinates, perform 6D sensor fusion, potentially on-chip and for example according to the algorithm described in [21], and report their orientations, described by the quaternions $\mathcal{S}_1^1 \mathbf{q}$ and $\mathcal{S}_2^2 \mathbf{q}$. We can easily calculate the body segment orientations $\mathcal{B}_i^i \mathbf{q} = \mathcal{S}_i^i \mathbf{q} \otimes \mathcal{S}_i^i \mathbf{q}$, $i = 1, 2$, with $\otimes$ denoting quaternion multiplication.

Without the use of magnetometers, the absolute heading of each sensor is unknown, which can be described by the orientations being estimated in two different reference frames, $\mathcal{E}_1$ and $\mathcal{E}_2$. At each moment in time, the difference between the reference frames $\mathcal{E}_1$ and $\mathcal{E}_2$ is only a rotation around the vertical axis, which has the coordinates $[0 \ 0 \ 1]^T$ in both $\mathcal{E}_1$ and $\mathcal{E}_2$. Let $\delta(t)$ be the angle of this rotation, then the vector representation of the corresponding quaternion is

$$\mathcal{E}_1^2 \mathbf{q}(\delta) = [\cos(\frac{\delta}{2}) \ 0 \ 0 \ \sin(\frac{\delta}{2})]^T. \quad (1)$$

In 6D sensor fusion, i.e. without a common reference for the heading, $\delta(t = 0)$ is unknown. Even if $\delta(t = 0)$ would be known, for example from a known initial calibration pose, this knowledge soon loses its value, because $\delta(t)$ drifts slowly due to non-zero gyroscope bias in both IMUs.

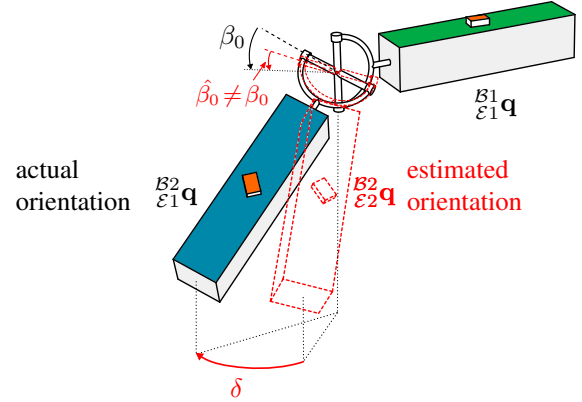


Fig. 2. Orientations of segments $\mathcal{B}_1$ and $\mathcal{B}_2$. Calculating the relative orientation between $\mathcal{B}_1^1 \mathbf{q}$ and $\mathcal{B}_2^2 \mathbf{q}$ without accounting for the difference in heading δ leads to a wrong joint orientation and wrong joint angles.

A. Orientation-based kinematic constraint for 2-DoF joints

Calculating relative segment orientations and joint angles directly from $\mathcal{B}_1^1 \mathbf{q}$ and $\mathcal{B}_2^2 \mathbf{q}$ leads to false results, cf. Fig. 2. Instead, we must first determine an estimate $\hat{\delta}(t)$ of the heading difference $\delta(t)$ and compensate it by bringing both body orientations into a common reference frame, i.e.

$$\mathcal{B}_2^2 \mathbf{q} = \mathcal{B}_1^1 \mathbf{q}^{-1} \otimes \mathcal{B}_2^2 \mathbf{q} = \mathcal{B}_1^1 \mathbf{q}^{-1} \otimes \mathcal{E}_1^2 \mathbf{q}(\hat{\delta}(t)) \otimes \mathcal{B}_2^2 \mathbf{q}. \quad (2)$$

Note that any estimate $\hat{\delta}(t)$ yields a relative orientation $\mathcal{B}_2^2 \mathbf{q} = [q_0 \ q_1 \ q_2 \ q_3]^T$. The intrinsic $z\text{-}x'\text{-}y''$ Euler angle

decomposition of this relative orientation is

$$\hat{\alpha} = \text{atan2}(2q_0q_3 - 2q_1q_2, q_0^2 - q_1^2 + q_2^2 - q_3^2), \quad (3)$$

$$\hat{\beta}_0 = \arcsin(2q_0q_1 + 2q_2q_3), \quad (4)$$

$$\hat{\gamma} = \text{atan2}(2q_0q_2 - 2q_1q_3, q_0^2 - q_1^2 - q_2^2 + q_3^2). \quad (5)$$

According to the definitions of $\mathcal{B}_1$ and $\mathcal{B}_2$ above, $\hat{\alpha}$ and $\hat{\gamma}$ are estimates of the joint angles, corresponding to rotations around $\mathbf{j}_1$ and $\mathbf{j}_2$, respectively, and the angle $\hat{\beta}_0$ corresponds to the fixed angle between $\mathbf{j}_1$ and $\mathbf{j}_2$. In general, a false $\hat{\delta}(t) \neq \delta(t)$ leads to a false $\hat{\beta}_0 \neq \beta_0$, cf. Fig. 2. This deliberation leads to the orientation-based constraint

$$\arcsin(2q_0q_1 + 2q_2q_3) - \beta_0 = 0, \quad (6)$$

which can be exploited to find the true heading difference $\delta(t)$ without using any magnetometer readings.

B. Singularity detection

The kinematic constraint (6) becomes singular, i. e. it yields no relative-heading information, if the segment orientations are such that one of the joint axes is perfectly vertical. In such a case, a change of relative heading and a rotation around the vertical joint axis cannot be distinguished, and the constraint is perfectly fulfilled for arbitrary values $\hat{\delta} \in \mathbb{R}$.

Due to inaccuracies in the estimated sensor orientations, this singularity will already deteriorate the results if any of the joint axes is close to vertical. In order to obtain a measure for how close the axes are to vertical, we transform the joint axes to the global coordinate system²

$$\mathbf{j}_1^{\mathcal{E}1}(t) = \mathbf{B}_1^1 \mathbf{q}(t) \otimes \mathbf{j}_1^{\mathcal{B}1} \otimes \mathbf{B}_1^1 \mathbf{q}(t)^{-1} =: [j_{1x} \ j_{1y} \ j_{1z}]^T, \quad (7)$$

$$\mathbf{j}_2^{\mathcal{E}2}(t) = \mathbf{B}_2^2 \mathbf{q}(t) \otimes \mathbf{j}_2^{\mathcal{B}2} \otimes \mathbf{B}_2^2 \mathbf{q}(t)^{-1} =: [j_{2x} \ j_{2y} \ j_{2z}]^T, \quad (8)$$

and define the *sample rating* $r(t)$ as the product of the lengths of their horizontal-plane projections (cf. Fig. 3):

$$r(t) := \|\mathbf{j}_{1,\text{proj}}^{\mathcal{E}1}\| \|\mathbf{j}_{2,\text{proj}}^{\mathcal{E}2}\|, \quad (9)$$

$$\|\mathbf{j}_{1,\text{proj}}^{\mathcal{E}1}\| = \|[j_{1x} \ j_{1y}]^T\|, \quad \|\mathbf{j}_{2,\text{proj}}^{\mathcal{E}2}\| = \|[j_{2x} \ j_{2y}]^T\|,$$

which goes to zero if any of the axes gets close to vertical.

²For quaternion multiplication, we implicitly regard three-dimensional vectors as quaternions with zero real part.

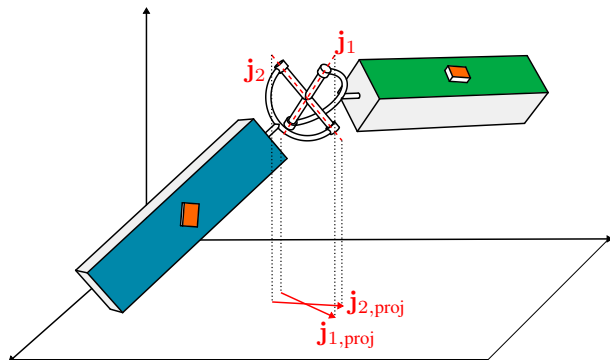


Fig. 3. Projections of the joint axes $\mathbf{j}_1$ and $\mathbf{j}_2$ in the horizontal plane.

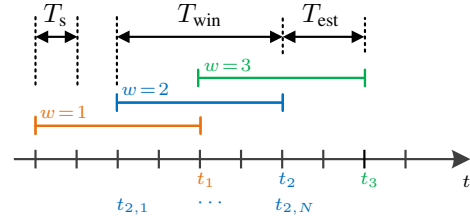


Fig. 4. Graphical representation of the estimation windows, the estimation time instants t_w , the sample time instants $t_{w,k}$ and the durations T_s , T_{est} , T_{win} .

C. Optimization-based estimation of the heading error

Instead of using the constraint sample-by-sample to estimate $\delta(t)$, we propose a window-based estimation over multiple samples. This is done to account for measurement inaccuracies and for the fact that biological joints only approximately fulfill the 2-DoF joint constraint.

At regular time intervals T_{est} , we perform an estimation of $\delta(t)$ based on data in a time window immediately preceding the estimation time. We denote the corresponding quantities with an index $w \in \mathbb{N}^+$, i.e. the estimation time instants are

$$t_w = wT_{\text{est}}. \quad (10)$$

Each window consists of N samples with a sample index $k = 1, \dots, N$, taken at a regular sampling interval T_s (which might be larger than the sampling rate of the sensor at which sensor fusion is performed) at the sampling times

$$t_{w,k} = wT_{\text{est}} + (k - N)T_s, \quad (11)$$

leading to a window duration of $T_{\text{win}} := T_s N$.

This implies that only previous data is used to estimate $\delta(t)$, making this approach suitable for use in realtime applications. See Fig. 4 for a visual representation of the estimation time windows.

Let $\mathbf{B}_1^2 \mathbf{q}(t_{w,k}) =: [q_0 \ q_1 \ q_2 \ q_3]^T$ be the relative orientation that (2) yields for a given $\hat{\delta}$. For each sample of the window, we define the weighted constraint error

$$e_{w,k}(\hat{\delta}) = (\arcsin(2q_0q_1 + 2q_2q_3) - \beta_0) r_{w,k} \quad (12)$$

with $k = 1, \dots, N$ and $r_{w,k} := r(t_{w,k})$.

As the heading difference $\delta(t)$ only changes slowly over time, we assume that it is constant during each window and find the angle δ_w that minimizes the sum of squares of the errors given in (12) over the window w :

$$\delta_w = \arg \min_{\hat{\delta}} \sum_{k=1}^N e_{w,k}(\hat{\delta})^2. \quad (13)$$

We determine analytical expressions for the Jacobian

$$[\mathbf{J}_w]_k = \frac{\partial e_{w,k}}{\partial \delta} \in \mathbb{R}^{N \times 1} \quad (14)$$

and use a Gauss-Newton algorithm to minimize the error for each window w , while using the estimate of the previous window as the initial value for each new window.

D. Singularity treatment and heading filter

The sample rating $r_{w,k}$ allows us to focus on the parts of a window that contain most relative-heading information. However, if the segments stay close to the singularity for durations that exceed the window duration T_{win} , the estimate of the corresponding windows might be wrong. To mitigate this, we first introduce the *window rating*

$$r_w := \sqrt{\frac{1}{N} \sum_{k=1}^N r_{w,k}^2}. \quad (15)$$

Then we design a filter that

- smoothes the δ_w trajectory, taking the rating into account
- extrapolates linearly if the rating is below a certain threshold
- does not introduce a delay if δ_w changes linearly.

To achieve this, we determine a filtered estimate $\delta_{f,w}$ by applying the following nested adaptive filter that estimates the heading bias b_w and uses it to extrapolate the heading difference whenever the rating is below a certain threshold:

$$b_w = b_{w-1} + s_w k_{b,w} (\delta_w - \delta_{w-1} - b_{w-1}) \quad (16)$$

$$\delta_{f,w} = \delta_{f,w-1} + b_w + s_w k_{\delta,w} (\delta_w - \delta_{f,w-1} - b_w) \quad (17)$$

$$s_w = \begin{cases} r_w & r_w \geq r_{\min} \\ 0 & \text{else} \end{cases} \quad (18)$$

with a rating threshold $r_{\min} \in [0, 1]$ and filter gains

$$k_{b,w} := \max \left(1 - \exp \left(-\ln 2 \frac{T_s}{\tau_b} \right), \frac{1}{w} \right), \quad (19)$$

$$k_{\delta,w} := \max \left(1 - \exp \left(-\ln 2 \frac{T_s}{\tau_\delta} \right), \frac{1}{w} \right). \quad (20)$$

τ_b and τ_δ are tunable half-error time constants for the bias and heading filter, respectively. The filter gains are increased for small w to ensure fast initial convergence.

E. Heading correction

We finally use (1), (2) and the filtered estimate $\delta_f(t)$ to determine the correction quaternion ${}^{\mathcal{E}_1^2}\mathbf{q}(t)$, the orientation of the second segment $\mathcal{B}_2$ in the reference frame of the first segment $\mathcal{E}_1$, and the relative orientation between both segments. The resulting quaternions can be used for joint angle calculation and for 3D visualization.

III. EXPERIMENTAL VALIDATION

The proposed method is validated experimentally in the metacarpophalangeal joints (MCP) of the index and middle fingers of the human hand. We design an experiment that allow us to obtain an approximate ground-truth reference for the heading difference δ and to investigate long-time stability of the obtained estimates during time motions with and without sufficient excitation, i.e. during motions that yield sufficient relative-heading information and during motions for which the segments remain close to the singularity.

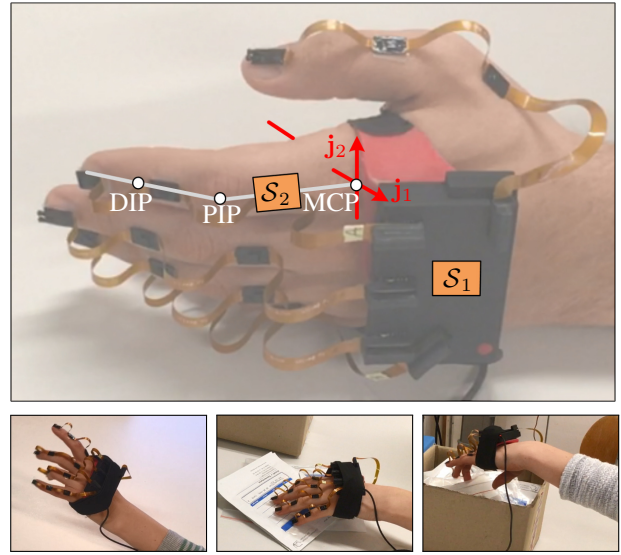


Fig. 5. Experimental setup and example scenes from the conducted experiments.

A. Setup

To measure the 6D orientation of the finger segments, we use a recently developed modular finger and hand motion capturing system [9] as shown in Fig. 5. One inertial sensor is attached to the back of the hand and tracks the movements of the metacarpal bones of both fingers. The second and third sensor are attached to the proximal phalanges of the index and middle finger. The MCP joints, which connect the proximal finger segments to the metacarpals, are approximate 2-DoF joints. The segment-to-sensor attachment is known by precise attachment. The IMUs measure the angular velocity and the acceleration at a rate of 100 Hz, and 6D sensor fusion is performed using the algorithm presented in [21]. The *abduction axis* $\mathbf{j}_1$ of the MCP is fixed in the coordinate system of the back of the hand. The *flexion axis* $\mathbf{j}_2$ is fixed in the coordinate system of the proximal phalange. Since the axes are approximately perpendicular we assume that $\beta_0 = 0$.

B. Obtaining an approximate reference

For validation, we define a reference pose that is taken several times during the experiment: The hand is lying flat on a horizontal surface with all fingers straight. For that pose the coordinate systems of all three sensors are approximately aligned. Therefore, whenever that pose is taken, the reference value δ_{ref} is determined as the heading difference that minimizes the rotation angle (maximizes the real part) of the relative quaternion ${}^{\mathcal{B}_1^2}\mathbf{q}$ in (2). Assuming that the heading drift is slow and approximately linear, we linearly interpolate δ_{ref} between the reference time instants.

Since this method relies on precise attachment of the sensors as well as exact positioning of the segments during resting phases, it yields only an approximate ground-truth reference, which we will only use for qualitative comparison and to examine whether the relative heading estimates $\delta_f(t)$ of the proposed method are long-time stable.

C. Conducted experiments

Two long-time experiments were conducted. The first experiment with a total duration of 8 min 41 s was designed with distinct movement phases to show phases with natural movement and longer phases with rest near the singularities. See Table I for an overview of the different movement phases and their duration.

TABLE I
OVERVIEW OF MOVEMENT PHASES FOR THE FIRST EXPERIMENT

Phase	Description	Start	End	Duration
M1	normal movement	0 s	90 s	90 s
M2	slow movement	90 s	153 s	63 s
M3	normal movement	153 s	247 s	94 s
M4	no movement, j_2 vertical	247 s	290 s	43 s
M5	normal movement	290 s	382 s	92 s
M6	no movement, j_1 vertical	382 s	424 s	42 s
M7	normal movement	424 s	521 s	97 s

The motion phases were chosen to include phases with and without excitation. Phases with movement are expected to yield good estimation results (M1, M3, M5, M7). During phases with approximately vertical axes (M4, M6), the data contains almost no heading information, but the singularity treatment as described in Section II-D should mitigate this.

The second experiment consists of functional movements like counting, typing, moving objects in and out of a box to test the method in real world applications. The second experiment has a total duration of 3 min 26 s. Videos of the performed motions can be found at [22].

IV. RESULTS AND DISCUSSION

For brevity, we restrict the detailed analysis in this paper to the first experiment and the index finger. More detailed results for both fingers and both experiments can be found at [22].

The settings for the estimation are chosen as follows: Every $T_{\text{est}} = 1$ s we analyze a window with a duration of $T_{\text{win}} = 20$ s, consisting of $N = 100$ samples taken every $T_s = 0.2$ s. We choose the filter time constants as $\tau_b = 15$ s and $\tau_\delta = 30$ s, with a rating threshold of $r_{\text{min}} = 0.4$.

A. Cost function analysis

To analyze the properties of the optimization problem for the different motion phases, we evaluate the cost function in (13) for each time instant t_w and for many evenly spaced angles $\delta \in [0^\circ, 360^\circ]$. Fig. 6 shows the cost function for the index finger over the entire experiment. It can be seen that a distinct global minimum exists for phases with movement, while the cost function becomes flat and exhibits a distinct second local minimum for phases without movement and a vertical joint axis (M4 and M6). During those phases no relative-heading information is available, and the minima do not reflect the true value of δ . From the overall trend, it can be seen that the minimum drifts slowly with an approximately constant slope of $0.6 \frac{^\circ}{\text{s}}$.

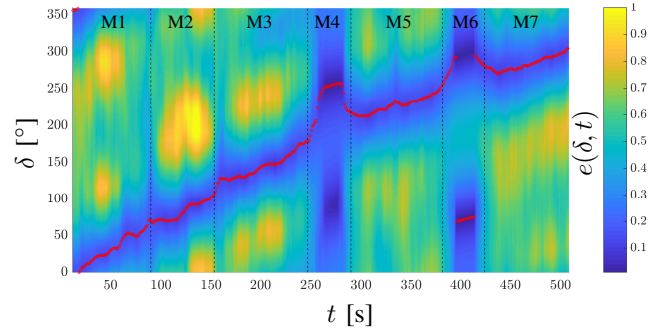


Fig. 6. Cost function for the index finger during the experiment. The minimum for each time instant t_w , i.e. the unfiltered heading estimate, is marked with a red dot.

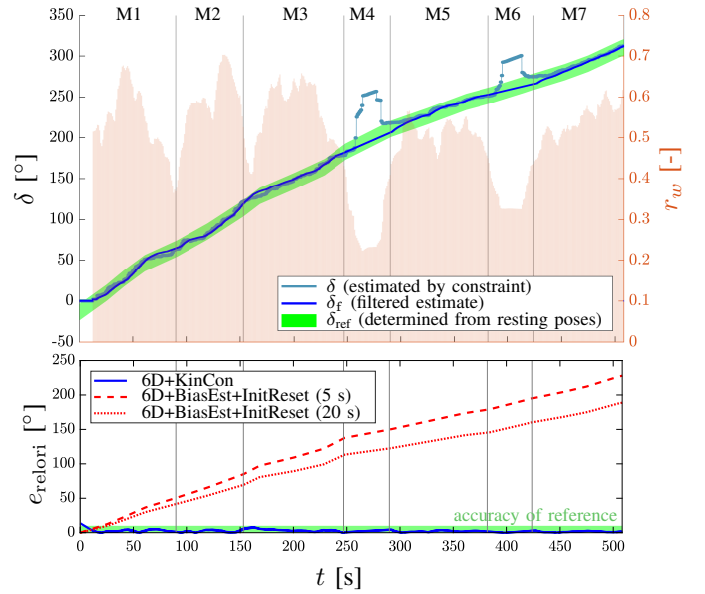


Fig. 7. Top: Estimated values for $\delta(t)$ and $\delta_f(t)$ as well as the reference $\delta_{\text{ref}}(t)$ (left axis) and window rating r_w (right axis). Bottom: Relative orientation error for the proposed method and the method described in [9].

B. Realtime heading error estimation

For realtime estimation of $\delta(t)$, the proposed algorithm is applied to the recorded data, i.e. for the estimation at a given time instant t_w only data from the past is used according to (13). The time series of the estimated $\delta(t)$ and $\delta_f(t)$ as well as the approximate ground-truth $\delta_{\text{ref}}(t)$ are shown in Fig. 7. The reference is depicted with a band of $\pm 10^\circ$ to indicate the aforementioned approximate accuracy of the reference value. It can be seen that during the phases with low window rating r_w , the unfiltered $\delta(t)$ diverges from the reference. However, the filtered estimate $\delta_f(t)$ remains close to the true value, even for long periods of time near the singularities.

C. Alternative 6D method

We additionally evaluate the 6D method proposed in [9]. During a time window without movement at the start of the measurement, the gyroscope readings are averaged to obtain an estimate of the gyroscope bias. After removing this bias, 6D sensor fusion is performed. At the beginning of the measurement, an initial-pose reset is performed, which

assumes a known pose to determine and correct a fixed heading offset. However, it is an inevitable fact that the resulting orientations are only valid for a limited time until they are deteriorated by integration drift resulting from residual and time-varying gyroscope bias. To obtain long-time stable motion tracking, the subject would have to repeat the initial reset procedure at least once per minute.

D. Error of the relative orientation

As explained above, accurate relative heading estimation is crucial for obtaining accurate relative segment orientations, which are required to determine joint angles and forward kinematics. We use $\delta_{\text{ref}}(t)$ to determine the approximate true relative orientation for each MCP joint and compare it to the relative segment orientation that is estimated by the proposed method. Denote this error by $e_{\text{reori}}(t)$.

Fig. 7 shows $e_{\text{reori}}(t)$ for the method presented in [9] (*6D+BiasEst+InitReset*) with bias estimation windows of 5 s and 20 s and for the new method based on kinematic constraints (*6D+KinCon*). As expected, when using initial reset and bias estimation, e_{reori} slowly drifts, while the proposed constraint-based method yields accurate results throughout the entire measurement.

V. CONCLUSION

We proposed and evaluated a novel method for magnetometer-free inertial motion tracking for joints with two degrees of freedom. The method uses an orientation-based constraint to determine the relative heading between the sensor orientations. A window-based optimization scheme was introduced, which renders the method real-time-capable. Furthermore, we examined a method for rating the reliability of the optimization result and proposed an adaptive filter that takes this rating into account and extrapolates the estimate when no relative-heading information is available.

With the proposed method it is possible to track the relative orientations and joint angles of joints with two degrees of freedom in real-world applications with realistic magnetic environments. While previous research is often limited to short-time analysis of rich motions, we demonstrated that the proposed method yields long-time stable results and is capable of handling phases in which no movement is performed or in which the movement does not contain heading information. Furthermore, in contrast to previously proposed methods, the current method works on arbitrary motions without the need to perform an initial reset with a predefined pose and without the need to determine sensor-to-joint position parameters.

Future work will focus on using the introduced constraint to simultaneously perform joint axis estimation in addition to relative heading estimation to facilitate plug-and-play magnetometer-free motion tracking.

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